

Octol

Octol is a melt-castable, high explosive mixture consisting of HMX and TNT in different weight proportions.

Composition

Two formulations are commonly used:

- 70% HMX & 30% TNT
- 75% HMX & 25% TNT

Given that HMX has a much higher detonation velocity than TNT (over 2,000 metres per second faster) and forms the main part of this explosive blend, the brisance characteristics of Octol can be inferred.

Applications

The applications of Octol are generally military; e.g., shaped charges and warheads that are used in guided missiles and submunitions. Octol is somewhat more expensive than RDX-based explosives, such as Composition B and Cyclotol. The advantage of Octol is that it significantly reduces the size and weight of the explosive charge required. These are important considerations where smart weapons such as guided missiles are concerned. A light (but effective) warhead means a superior power to weight ratio. This in turn results in a higher velocity missile with a longer range and shorter flight time. As a result, the target has less opportunity to recognise and evade the attack.

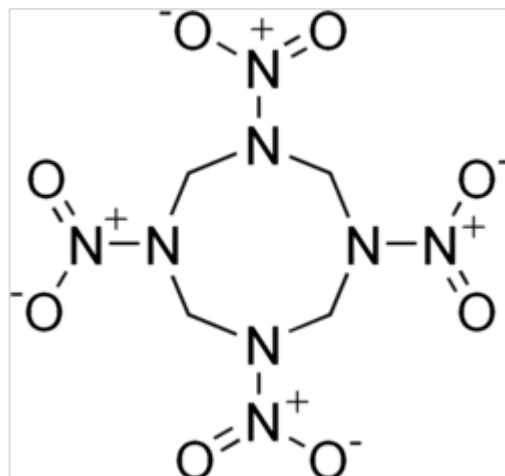
Alternative names

Octol is also referred to as **Oktol**, particularly in Eastern Europe.

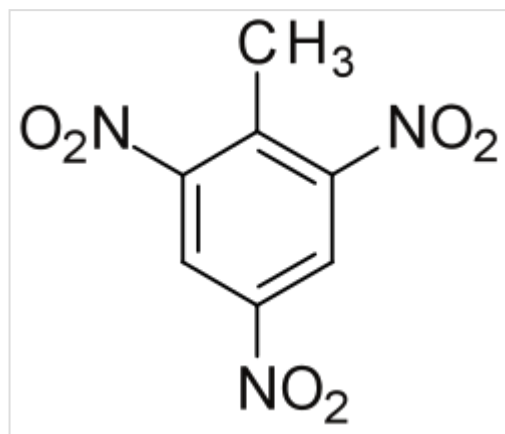
Nominal Properties

Reported properties:^[1]

- Density: 1.80 g/cm³
- Detonation Velocity: 8.55 mm/μsec



HMX



TNT

See also

- OKFOL, another HMX-based explosive.
- Cyclotol
- RE factor

References

1. "Detonation pressure measurements in TNT and octol" (<https://web.archive.org/web/20250202082337/https://apps.dtic.mil/sti/tr/pdf/AD0802251.pdf>) (PDF). Archived from the original (<https://apps.dtic.mil/sti/tr/pdf/AD0802251.pdf>) (PDF) on 2025-02-02.

External links

- Reference to Octol used as a booster (<http://www.globalsecurity.org/wmd/ops/minor-uncle.htm>)

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